2D-PDM: three-stream RGB-D framework

Single-view scene observation

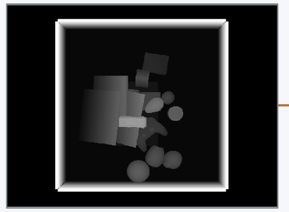
1. Inputs

Scene RGB I



640 × 480

Observed depth D



Registered to RGB

Target reference



RGB thumbnail + text



Reference mask

Text prompt

2a. Similarity stream

Trained; frozen for fusion

DINOv3 scene + target RGB (F)

Four scene feature layers

target

SigLIP target RGB + text (F)

Masked pooling + semantic adapter (T)

reference mask

Multi-layer matching + feature CNN (T)

F_S

64×30×40

1×1+σ

S map



2b. Occlusion stream

Trained; frozen for fusion

DINOv3 scene + target RGB (F)

Four scene feature layers

Target: full-frame mean pooling

Depth ResNet18 depth + valid mask (T)

Global FiLM geometry 68 + MLP (T)

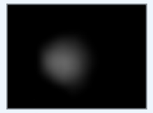
Multi-layer matching + feature CNN (T)

F_O

64×30×40

1×1+σ

O map



workspace mask

2c. Complexity stream

Trained; frozen for fusion

DINOv3 RGB last-layer features (F)

Layer 11 features

RGB CNN (T)

MultiMAE observed-depth encoder (F)

Depth encoding; no depth prediction

Projection + concat dilated CNN (T)

F_C

64×30×40

Softplus

C map



3. Feature fusion

SOC: trained 13 epochs; best 10

Concatenate stream features

$[F_S; F_O; F_C]$

192 × 30 × 40

1×1 Conv: 192 → 64
GN8 + GELU

3×3 Conv: 64 → 64
GN8 + GELU
× 2 blocks

1×1 Conv: 64 → 1
logits

σ: 1 × 30 × 40

Prediction **Amodal GT**



Actual output Native mask

BCEWithLogits
GT = avg16(amodal mask)

One learned CNN; no attention

Individual training Separate supervision with similarity, occlusion and regional-complexity GT.

F Frozen pretrained backbone **T** Module learned in its stream Feature tiles are schematic; map thumbnails are actual test predictions.

Fusion: all S/O/C modules frozen. One shared scene DINO forward; repeated nodes show logical uses.

SOC best10: known16 test (24,000)

Amodal peak hit: 85.06%

Amodal mIoU: 50.92%